

2.4 inch LCD TFT

(2.4 英寸真彩色液晶显示屏)

240RGBx320 Resolution and 65K color

(240x320 分辨率, 6 万 5 千种色)

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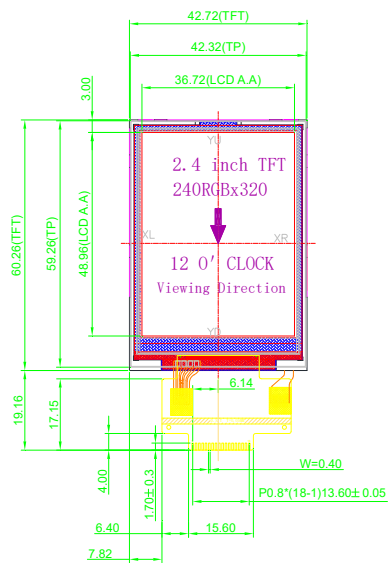
1.General Description 基本描述

MODEL NO 产品型号	ZJY240IT010
Display Mode 显示模式	Transmissive 全透
Display Format 显示格式	Graphic 240RGB*320 Dot-matrix 240xRGBx320 图形点阵
Input Data 显示屏接口类型	4 Line-SPI interface 4 线-SPI 串口
Viewing Direction 视角方向	12 o'clock 12 点钟
Drive 显示屏驱动芯片	ILI9341

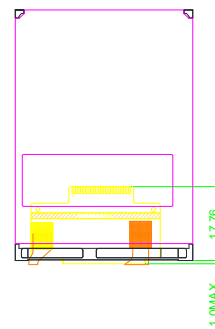
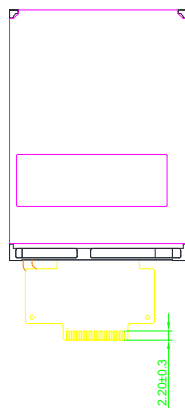
2. Mechanical Specification 机械规格

Item	Specifications	Unit
Dimensional outline 显示屏外围尺寸	42.72(W)*60.26(H)*2.60(T) (不带触摸) 42.72(W)*60.26(H)*3.80(T) (带触摸) (FPC not include)	mm
Resolution 分辨率	240RGB*320	dots
LCD Active area 显示尺寸	36.72(W)*48.96 (H)	mm
Pixel size 像素尺寸	0.153(W)*0.153(H)	mm

3.Mechanical Dimension 机械尺寸图



带触摸3.80MAX(不含双面胶厚度)
 不带触摸2.60MAX(不含双面胶厚度)



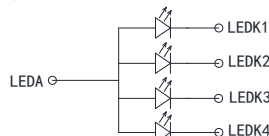
弯折示意图
 展开出货

PIN	SYMBOL
1	GND
2	RESET
3	SCL
4	D/C
5	CSX
6	SDA
7	SDO
8	GND
9	VDD
10	LEDA
11	LEDK1
12	LEDK2
13	LEDK3
14	LEDK4
15	XL(X-)
16	YU(Y-)
17	XR(X+)
18	YD(Y+)

NOTES:

1. DISPLAY TYPE:TFT
2. OPERATING TEMP: $-20^{\circ}\text{C}\sim 70^{\circ}\text{C}$
3. STORAGE TEMP: $-30^{\circ}\text{C}\sim 80^{\circ}\text{C}$
4. LCD DRIVER: COG(IC:ILI9341);
5. BACKLIGHT: 4 CHIP-WHITE LED(Parellel)
6. GENERAL TOLERANCE: ± 0.20
7. ROHS

背光电路图:



CIRCUIT DIAGRAM
 (I=60MA V=3.0-3.4V)

郑州中景园电子技术有限公司 Zhengzhou Zhongjingyuan Electronic Technology Co., LTD				第三视角:
产品型号:	ZJY240IT010			
部品型号:	LCM外形图	确认:		
版本:	v0.0	单位:	MM	审核:
日期:	2016.3.19	比例:	1:1	设计:
UNMRKED TOLERANCE: ± 0.20 mm		页码:	1 OF 1	

4. Electrical Maximum Ratings 电气极限

Item 项目	Symbol 符号	Min 最小值	Max 最大值	Unit 单位	Note 备注
Supply voltage (VDDI) 工作电压(VDDI)	V	1.8	3.3	V	-
Supply voltage (VDD) 工作电压(VDD)	V	2.8	3.3	V	-
Operating temperature 工 作温度范围	T _{OPR}	-20	70	℃	-
Storage temperature 存储温度范围	T _{STR}	-30	80	℃	-

※NOTE: VDDI 和 VDD 可以直接连一起, 共用一组 (2.8V~3.3V) 电压供电。

5. Brightness characteristic&Power dissipation 亮度特性&功耗

Item 项目	Symbol 符号	Min 最小值	Typical 典型值	Max 最大值	Unit
LED module Forward voltage LED 背光源正向电压	V _{LED}	2.9	3.1	3.3	V
LED module current LED 背光源电流	I _{LED}	-	60	-	mA
LCD Surface Luminance 显示屏表面亮度	L _S	250	300	-	Cd/m ²
LCM Surface brightness uniform LED 背光源均匀度	L _D	80	-	-	%
LCD power dissipation 显示屏总功耗	P _{LCD}	-	0.22	-	W

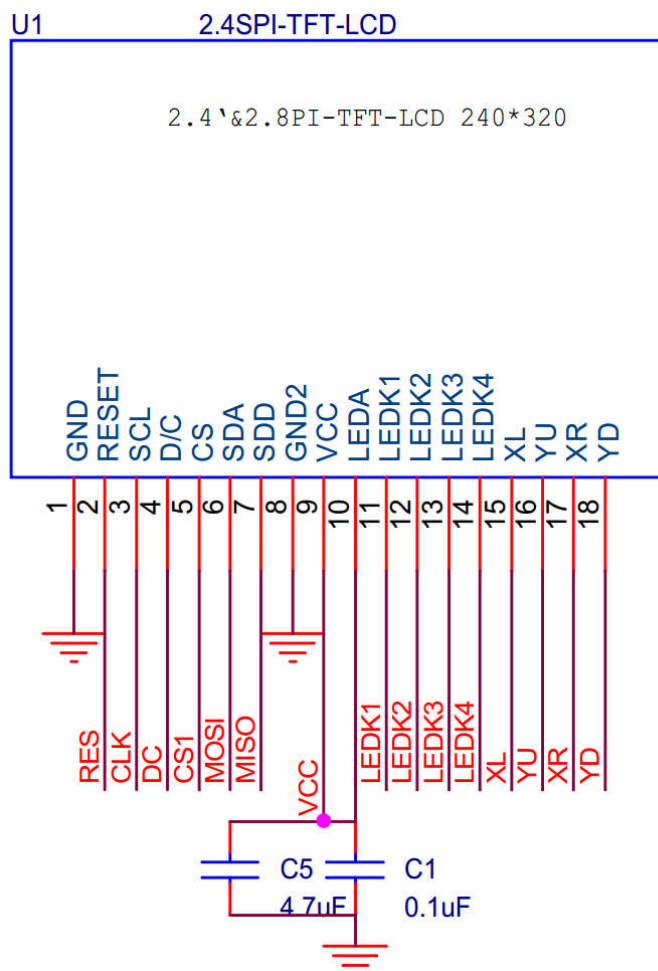
※NOTE: $P_{LCD} = VDD * (I_{LED} + I_{LCD})$

6. Module Function Description 显示屏脚位定义

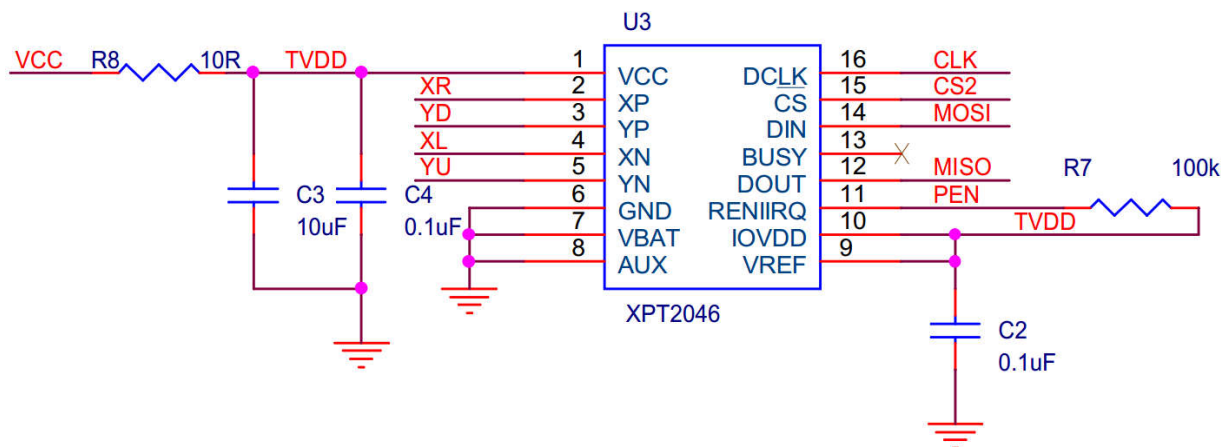
PIN No. 引脚序号	Symbol 引脚名称	Description 作用描述	Notes 备注
1	GND	Ground (接地脚)	-
2	RESET	-This signal will reset the device and it must be applied to properly initialize the chip. -Signal is active low. (显示屏复位脚, 低电平有效)	-
3	SCL	-This pin is used to be serial interface clock. (4 线-SPI 串口时钟)	-
4	D/C	- Display data/command selection pin in 4-line serial interface. (4 线-SPI 数据/指令选择选择脚)	-
5	CSX	-Chip selection pin Low enable. High disable. (显示屏驱动芯片选脚, 低电平使能)	-
6	SDA	SPI interface input pin. -The data is latched on the rising edge of the SCL signal. -If not used, please fix this pin at VDDI or DGND level. (4 线-SPI 数据输入, 不用时接 VDDI 或 GND)	-
7	SDO	-SPI interface output pin. -The data is output on the falling edge of the SCL signal. -If not used, let this pin open. (4 线-SPI 数据输出, 不用时悬空)	-
8	GND	Ground (接地脚)	-
9	VDD	Power Supply for Analog, Digital System and Booster Circuit. (显示屏主电源供电脚 2.8-3.3V)	-
10	LED A	Anode of Backlight (2.9V-3.3V Typical:3.1V) (背光正极供电脚, 电压范围:2.9-3.3V, 典型值:3.1V)	-
11	LED K1	Cathode of Backlight (背光负极供电脚)	-

12	LEDK2	Cathode of Backlight (背光负极供电脚)	-
13	LEDK3	Cathode of Backlight (背光负极供电脚)	-
14	LEDK4	Cathode of Backlight (背光负极供电脚)	-
15	XL	Touch panel Logical foot (四线电阻触摸屏逻辑脚)	-
16	YU	Touch panel Logical foot (四线电阻触摸屏逻辑脚)	-
17	XR	Touch panel Logical foot (四线电阻触摸屏逻辑脚)	-
18	YD	Touch panel Logical foot (四线电阻触摸屏逻辑脚)	-

附图 6-1: 显示屏 ZJY240IT010 线-SPI 串口参考应用电路



附图 6-2: 触摸屏参考应用电路

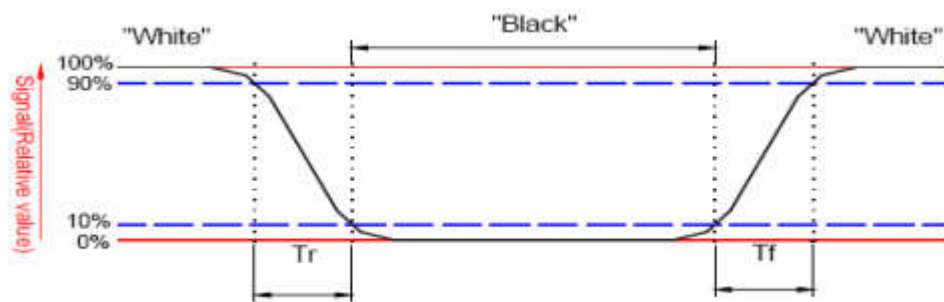


※NOTE:

- 1.若为不带触摸版本，参考附图 6-1 连接电路，把显示屏的 XL、YU、XR、YD 四个触摸脚悬空即可。
- 2.若为带触摸版本，参考附图 6-2 触摸应用连接电路。

7.Response time&Contrast ratio 响应时间与对比度

Item 项目	Symbol 符号	Condition 条件	Remark			Unit 单位
			Min. 最小值	Typ. 典型值	Max. 最大值	
Response time 响应时间	Tr+Tf	$\theta = 0^\circ$	-	25	40	ms
Contrast ratio 对比度	CR	$\theta = 0^\circ$	350	500	-	-



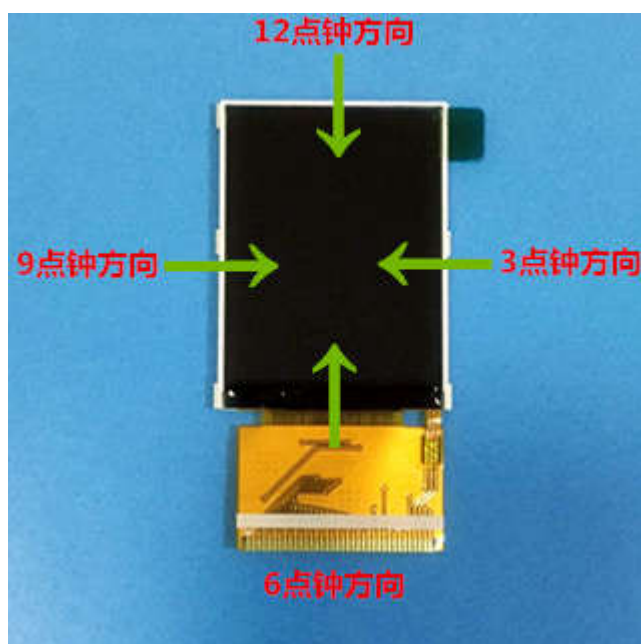
响应时间图示

$$\text{Contrast ratio (CR)} = \frac{\text{Brightness on the "white" state}}{\text{Brightness on the "black" state}}$$

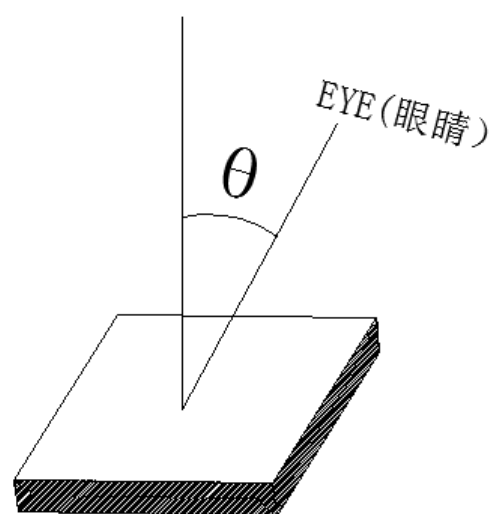
对比度计算公式

8.Viewing Angle 视角宽度

Item 项目	Symbol 符号	Condition 条件	Remark			Unit 单位
			Min. 最小值	Typ. 典型值	Max. 最大值	
Viewing angle 视角宽度	Top 12 点钟方向	$CR \geq 10$ 对比度大于等于 10	40	50	-	Deg. 度
	Bottom 6 点钟方向	$CR \geq 10$ 对比度大于等于 10	55	65	-	
	Left 9 点钟方向	$CR \geq 10$ 对比度大于等于 10	55	65	-	
	Right 3 点钟方向	$CR \geq 10$ 对比度大于等于 10	55	65	-	



垂直于屏表面



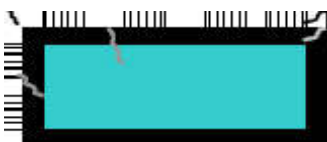
NOTE:3 点, 6 点, 9 点, 12 点方向视角的大小指的是垂直于屏表面的线眼睛视线之间的夹角(θ)。

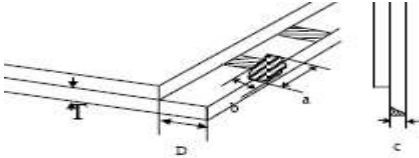
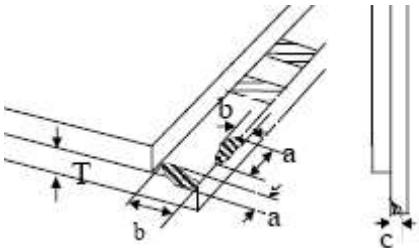
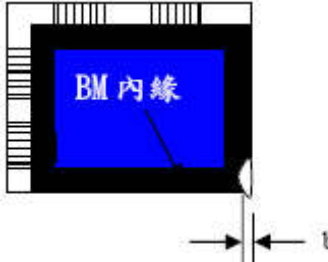
9. Reliability Trial 可靠性实验

NO. 序号	ITEM 实验项目	CONDITION 实验环境	CRITERION 实验规范
1	High Temperature Non-Operating Test 高温存储实验	80℃*120Hrs	No Defect Of Operational Function In Room Temperature Are Allowable 室温运行功能无缺陷
2	Low Temperature Non-Operating Test 低温存储实验	-30℃*120Hrs	
3	High Temperature/Humidity Non Operating Test 高温高湿实验	60℃*90%RH*120Hrs	
4	High Temperature Operating Test 高温工作实验	70℃*72Hrs	
5	Low Temperature Operating Test 低温工作实验	-20℃*72Hrs	
6	Thermal Shock Test 热冲实验	-20℃ (30Min) ↔ 70℃ (30Min) *10CYCLES	

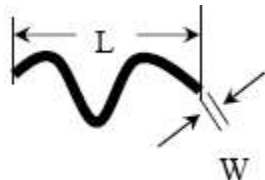
10. Inspection standards 检验标准

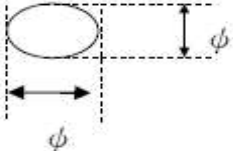
10.1 Glass defect

NO	Defect item	Criteria	Remark
1	Dimension Unconformity (Major defect)	By Engineering Drawing	
2	Cracks (Major defect)	1. Linear cracks panel 【Reject】 2. Nonlinear crack contrast by limited sample	
3	Glass extrude the conductive area (minor defect)	a: disregards and no influence assemblage. 1) $b \leq 1/3$ Pin width (non bonding area) 【Accept】 2) bonding area ≤ 0.5 mm 【Accept】	A: Length, b: Width
4	Pin-side ,conductive area damaged	(a c: disregards) $b \leq 1/3$ of effective length for	a: length, b: Width, c: Thickness

	(minor defect)	bonding electrode 【Accept】	
5	Pin-side, non-conductive area damaged (minor defect)	1) Damage area don't touch the ITO (Including contraposition mark, except scribing mark) 【Accept】 2) $C < T$ $b \leq BM1/3$ of width 【Accept】 3) $c = T$ b not touch the seal glue 【Accept】 4) a disregards	a: Length, b: Width c: Thickness 
6	Non-pin-side damage (minor defect)	$c < T$ 1) b exceeds $1/3 BM$ 【Reject】 $c = T$ b not touch the seal glue 【Reject】	c : Thickness b : width of  damage

10.2 LCD appearance defect (View area)

NO	Defect item	Criteria		Remark
1	Fiber、glass cratch、polarizer scratch/folded (minor defect)	Specification	Allowable	note1: L: Length, W: Width note2: disregard if out of AA 
		$W \leq 0.03\text{mm}$	disregard	
		$0.03\text{mm} < W \leq 0.05\text{mm};$ $L \leq 3.0\text{mm}$	2	
		$0.05\text{mm} < W \leq 0.1\text{mm};$ $L \leq 3.0\text{mm}$	1	
		$W > 0.1\text{mm}; L > 3.0\text{mm}$	0	
2	Polarizer bubble、concave and convex (minor defect)	$\phi \leq 0.2\text{mm}$	disregard	note1: $\phi = (L+W)/2$, L: Length, W: Width note2: disregard if out of AA
		$0.2\text{mm} < \phi \leq 0.3\text{mm}$	2	
		$0.3\text{mm} < \phi \leq 0.5\text{mm}$	1	
		$0.5\text{mm} < \phi$	0	
3	Black dots、dirty dots、impurities、eye winker (minor defect)	$\phi \leq 0.15\text{mm}$	disregard	note2: disregard if out of AA
		$0.15\text{mm} < \phi \leq 0.25\text{mm}$	2	
		$0.25\text{mm} < \phi \leq 0.3\text{mm}$	1	

		$0.3\text{mm} < \phi$	0	
4	Polarizer prick (minor defect)	$\phi \leq 0.1\text{mm}$	disregard	note1: $\phi = (L+W)/2$, L=Length, W=Width note2: the distance between two dots > 5mm
		$0.1\text{mm} < \phi \leq 0.25\text{mm}$	3	
		$\phi > 0.25\text{mm}$	0	

11.Package Method 包装方法

显示屏出货包装示意图：

